David Adeshina Arungbemi

I follow what moves me, then test whether it's true. Also writes as Suzume.

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RESEARCH INTERESTS

- What a model reports versus what drives its behaviour
- Training language models on stories
- Reading what a model is poised to say
- Brain and model responses to images
- Translation between kinds of knowledge

EDUCATION

B.Eng. Computer Engineering, Nile University of Nigeria 2019 – 2024

First Class Honours, CGPA 5.0 / 5.0. Valedictorian and best overall graduate. 90% scholarship. Thesis (group project, three students): a wrist-worn gesture-recognition device using motion sensors. Our approach scored 94.9% on the public 6DMG dataset; retrained on our own device, it reached about 60% with two new users.

PUBLICATIONS

Narrative Transportation and Speciesism in Large Language Models 2026

D. A. Arungbemi. Preprint, not peer-reviewed. Zenodo. doi:10.5281/zenodo.21294610 · PDF

I gave five language models a story about an animal, the same facts written plainly, or an essay arguing for animals. Only the story clearly made their answers more caring toward that animal.

An End to End Wearable Device and System for Indefinite, Continuous, Real Time Gesture Recognition of Directional and Shape-Based Arm Gestures 2025

IEEE Access, vol. 13, pp. 153436–153463. Peer-reviewed. Co-author.

doi:10.1109/ACCESS.2025.3602871

My part: the AI that recognises the gestures, and the interface of the live app people use with it.

FELLOWSHIPS & PROGRAMMES

Research Fellow, MARS 5.0, Cambridge AI Safety Hub 2026 – now

A mentored AI safety research programme. I test whether training a language model on a story changes the values it acts on, comparing each story with the same text rewritten neutrally or with its ending flipped. I design the experiments and datasets, run the training (supervised fine-tuning, distillation, GRPO) on Gemma and Qwen models, and present results to the cohort. Mentor: Elizaveta Tennant (Google DeepMind / UCL).

- **A story that acts without being named.** Trained on a fable written as its own answers about itself, a small model (Gemma 4) shifted its decisions the way the fable does when it is in the prompt, and took the deceptive option on MACHIAVELLI less often than the base model or a control; no reasoning trace named the fable.
- **Is the story in the workspace?** With Anthropic's Jacobian lens the fable showed plainly when it was in context; in the trained model its signal was matched by the control, so it came from the training, not the fable.
- **Describing versus using a reasoning process.** I trained a small model on first-person accounts of how it reasons: by projecting a known story onto a new situation (Mark Turner, *The Literary Mind*). It described the process almost every time it was asked and brought that story into its reasoning on about 70% of 2,000 dilemmas, yet its overall decisions barely changed: whether it learned the process or only its words is still open.

Research Fellow, Futurekind Fellowship, Electric Sheep 2026

A 12-week fellowship on AI and animal welfare, where I wrote the preprint above.

Selected, Diverse Intelligences Summer Institute 2026
An interdisciplinary programme on how humans, animals and machines think.

Finalist, MATS
Passed the coding and writing stages to reach the final round of a selective AI safety research programme, in its AI welfare stream.

EXPERIENCE

Co-founder & Lead AI, Gyronics Technologies Nov 2024 – now
An assistive wearable for accessibility that recognises arm gestures. I designed the device's look in Blender and built its gesture recognition, helped build the tool our team uses to watch participants' gesture data live, and made a view that draws a gesture as the path of a virtual pen.

Research Intern, National Centre for AI and Robotics, Abuja Feb 2025 – Jan 2026
A live video system that tracks people and objects and maps how near they are to each other, and a robot arm guided only by a camera.

SELECTED PROJECTS

Aesthetics Optimizer 2026
Predicting the brain's fMRI response (BOLD5000) to a new image, personalised from a person's ranked image history, with a desktop app that tunes generative art to raise the prediction. The predictions were weak (mean r ≈ 0.08), and the write-up analyses why.

DiverseIntelligence in progress
Can people from different fields solve one problem together? AI agents playing Feynman, Kierkegaard and William James work on quantum gravity while a translator rewrites each message so the listener can follow it; a physics-only group works without it, for comparison.

Narrative–Technical Embedding Space 2025
Contrastive learning to tell stories from technical writing, as a first step toward translating between them.

Codebook Genome 2025
A VQ-VAE turns each small image into a sequence of 64 codes, which I read like DNA: as a strand, a grid and a ring, three ways of seeing the same representation.

SELECTED WRITING

A Self-Reasoning System 2025
A sketch of Leah, a system that turns a problem into a structured representation and reasons by moving through it, with every step readable.

The Gap Between Sound and Meaning 2026
I asked two AIs to hear words as pure sound. They couldn't: for them there is no gap between sound and meaning.

17 articles in all at suzume1.medium.com

SKILLS

Research	Reading papers, designing and running experiments, evaluating results, writing them up and presenting them; fMRI modelling (public BOLD5000 data)
Models	Fine-tuning small language models (supervised, GRPO) with PyTorch, Hugging Face and Unsloth on RunPod; benchmarks; LLM-as-judge; multi-agent simulations
Building	Python (main) with Claude Code; C++ (ImGui / ImPlot); apps and visualisations in React and Three.js
Design	Wearable device design, data visualisation, Blender, creative coding in Processing